Wireshark Lab: Assignment 2

Advanced Network Security Lab Assignment: Wireshark HTTP Analysis

Objective: In this lab, you will use Wireshark to analyze HTTP traffic and understand the interactions between client and server. You will also explore HTTP message formats, authentication, and basic security mechanisms.

- **Task 1: Basic HTTP GET/Response Interaction**
- **Task 2: HTTP Authentication**

Assignments Instructions:

1- Please briefly justify/explain your approach and/or answers.

2 Provide **screenshots** for your answers captured in

Wireshark.

Having gotten our feet wet with the Wireshark packet sniffer in the introductory lab, we're now readyto use Wireshark to investigate protocols in operation. In this lab, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats, retrieving large HTML files, retrieving HTML files with embedded objects, and HTTP authentication and security.

1. **The Basic HTTP GET/response interaction.**

Let's begin our exploration of HTTP by downloading a very simple HTML file -one that is very short,and contains no embedded objects. Do the following:

1. Start up your web browser.
2. Start up the Wireshark packet sniffer, as described in the Introductory lab (but don't yet begin packet capture). Enter "http" (just the letters, not the quotation marks) in the display-filter-specification window, so that only captured HTTP messages will bedisplayed later in the packet-listing window. (We're only interested in the HTTP protocol here, and don't want to see the clutter of all captured packets).
3. Wait a bit more than one minute (we'll see why shortly), and then begin Wiresharkpacket capture.
4. Enter the following to your browser
   http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html
   Your browser should display the very simple, one-line HTML file.

5. Stop Wireshark packet capture.

Your Wireshark window should look similar to the window shown in Figure 1. If you are unable to run Wireshark on a live network connection, you can download a packet trace that was created when the steps above were followed.

Note-1: Download the zip file from canvas, file/Support Materials/Labs_HWs/wireshark-traces.zip and extract the file http- ethereal-trace-1. The traces in this zip file were collected by Wireshark running on one of the author's computers, whileperforming the steps indicated in the Wireshark lab. Once you have downloaded the trace, you can load it into Wireshark and view the trace using the *File* pull down menu, choosing *Open*, and then selecting the http-ethereal-trace-1 trace file. The resulting display should look similar to Figure 1. (The Wireshark user interface displays just a bit differently on different operating systems, and in different versions of Wireshark).

Note-2: You should ignore any HTTP GET and response for favicon.ico. If you see a reference

to this file, it is your browser automatically asking the server if it (the server) has a small icon file that should be displayed next to the displayed URL in your browser. We'll ignore references to this pesky file in this lab.

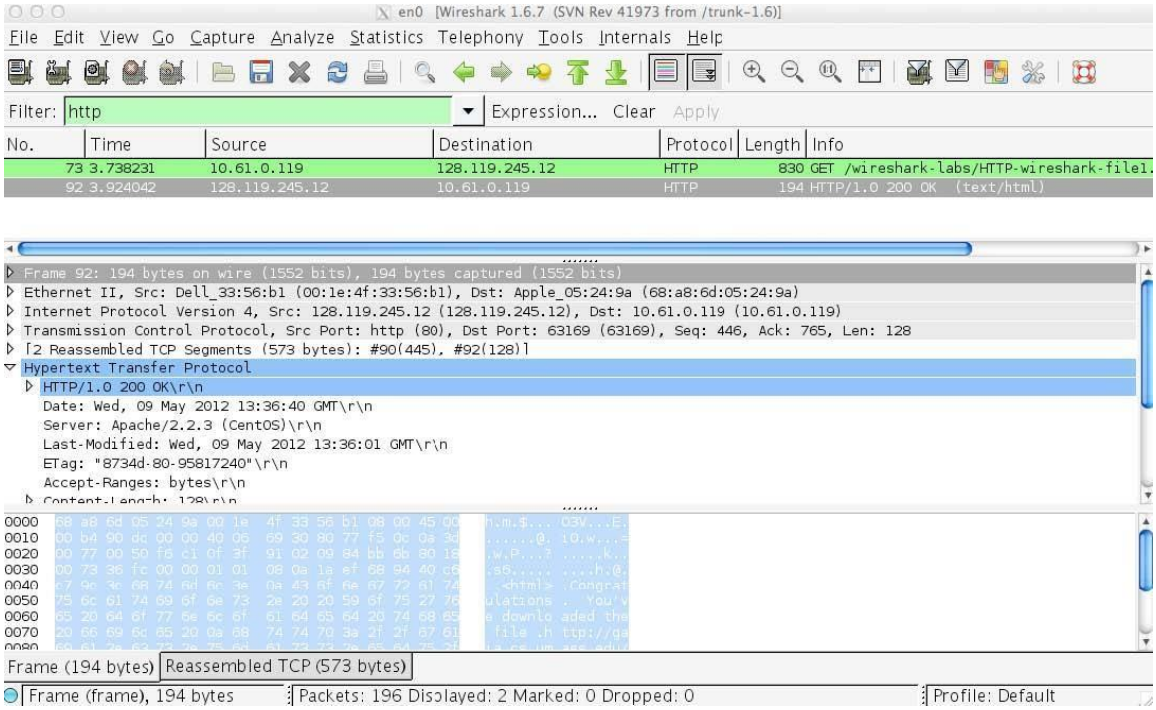

Figure 1: Wireshark Display after http://gaia.cs.umass.edu/wireshark-labs/ HTTP Wireshark-file1.html has been retrieved by your browser.

The example in Figure 1 shows in the packet-listing window that two HTTP messages were captured: the GET message (from your browser to the gaia.cs.umass.edu web server) and the response message from the server to your browser. The packet-contents window shows details of the selected message (in this case the HTTP OK message, which is highlighted in the packet-listing window). Recall that since the HTTP message was carried inside a TCP segment, which was carried inside an IP datagram, which was carried within an Ethernet frame, Wireshark displays the Frame, Ethernet, IP, and TCP packet information as well. We want to minimize the amount of non-HTTP data displayed (we're interested in HTTP here, and will be investigating these other protocols is later labs), so make sure the boxes at the far left of the Frame, Ethernet, IP and TCP information have a plus sign or a right-pointing triangle (which means there is hidden, un-displayed information), and the HTTP line has a minus sign or a down-pointing triangle (which means that all information about the HTTP message is displayed).

By looking at the information in the HTTP GET and response messages, answer the following questions. When answering the following questions, you should print out the GET and response messages and indicate wherein the message you've found the information that answers the following questions. When you hand in your assignment, annotate the output so that it's clear where in the output you're getting the information for your answer.

1. **Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the serverrunning?**
2. **What languages (if any) does your browser indicate that it can accept to the server?**
3. **What is the IP address of your computer and the IP address of the gaia.cs.umass.edu server?**
4. **What is the status code returned from the server to your browser?**
5. **When was the HTML file that you are retrieving last modified at the server?**
6. **How many bytes of content are being returned to your browser?**

In your answer to question 5 above, you might have been surprised to find that the document you just retrieved was last modified within a minute before you downloaded the document. That's because (for this particular file), the **gaia.cs.umass.edu** server is setting the file's last-modified time to be the current time, and is doing so once per minute. Thus, if you wait a minute between accesses, the file will appear to have been recently modified, and hence your browser will download a "new" copy of the document.

2. HTTP Authentication

Finally, let's try visiting a web site that is password-protected and examine the sequence of HTTP message exchanged for such a site. The URL http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html is password protected. The username is "wireshark-students" (without the quotes), and the password is "network" (again, without the quotes).So, let's access this "secure" password-protected site. Do the following:

- Make sure your browser's cache is cleared, as discussed above, and close down your browser. Then, start up your browser
- Start up the Wireshark packet sniffer
- Enter the following URL into your browser http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html
- Stop Wireshark packet capture, and enter "http" in the display-filter-specification window, so that only captured HTTP messages will be displayed later in the packet-listing window.
- (*Note:* If you are unable to run Wireshark on a live network connection, you can use the http-ethereal-trace-5 packet trace to answer the questions below; see footnote 2. This trace file was gathered while performing the steps above on one of the author's computers.)

Now let's examine the Wireshark output. You might want to first read up on HTTP authentication by reviewing the easy-to-read material on "HTTP Access Authentication Framework" at http://frontier.userland.com/stories/storyReader$2159

Answer the following questions:

7. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?
8. When your browser sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?